AI EMPLOYEE PLATFORM · DOCUMENT 02

System Design & Architecture

Technical architecture, data model, AI and voice design, APIs, security, infrastructure and delivery engineering for the AI Employee SaaS platform

One Platform → Multiple AI Employees → Multiple Industries

DOCUMENT
SDD-02 System Design & Architecture

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PREPARED FOR
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1. Introduction

This document is the technical design of the AI Employee platform. It describes how the system is structured, how data is modelled and isolated per company, how the AI and voice engines work, how APIs, frontends, security, infrastructure, testing and delivery are organised, and which decisions were taken and why. It is the blueprint engineering will follow phase by phase.

1.1 Scope

- Covers the MVP (phases P0-P8) in full detail and the post-MVP phases (P9-P12) at the level needed so the MVP architecture supports them without rework.
- Commercial items (timeline, cost, running cost) are in the separate commercial proposal.
- Provider names are candidates; final provider choice per language is made in the Phase 0 proof of concept.

1.2 Conventions

Term	Meaning
Company / tenant	A registered business. All its data carries company_id.
AI Employee	A configured agent (e.g., "Sara, Sales") owned by a company; a configuration, not a separate model.
Template	An industry template version: fields, catalogues, tools, statuses, scoring, instructions.
Role	An AI Employee type: Sales, Receptionist, Appointment, Follow-up, Support, Collection, custom.
Tool	A server-side action the AI may call, e.g. search_property, book_appointment.
Adapter	A thin interface wrapping one external provider so it can be swapped.
Feature IDs	IDs such as CRM-04 refer to the PRD-01 feature catalogue.

2. Goals, Non-Goals & Architectural Drivers

2.1 Goals

- Serve thousands of businesses on one multi-tenant platform with strict per-company isolation.
- One AI engine for every industry, role and channel; new industries are configuration, not code.
- Natural, low-latency voice conversations in Indian languages, including code-mixed speech.
- Everything configurable from Super Admin; no hard-coded business values.
- Replaceable AI, voice, telephony, messaging and payment providers.
- API-first: every capability exposed through versioned Django REST APIs built alongside each feature, so the web panels and a future Flutter mobile app (Android + iOS) use the same backend, data and workflows.
- Testable in phases, each passing automated tests, GitHub CI and security checks.

2.2 Non-Goals

- Training or hosting our own foundation LLM, STT or TTS models (we use providers through adapters).
- Unofficial WhatsApp automation or unauthorised robocalling.
- Integration with external CRMs or external calendars (own CRM and calendar by decision).
- AI Marketing Employee (pending client definition).

2.3 Quality Attributes and Targets

Attribute	Target / approach
Tenant isolation	Zero cross-tenant access. Three enforcement layers + automated isolation tests at every gate.
Voice latency	P50 ≤ 1.2 s, P95 ≤ 2.0 s end of speech → first AI audio (validated in P0).
Availability	99.9 % monthly for API and panels; voice call setup success ≥ 99.5 %.
API performance	P95 ≤ 400 ms reads, ≤ 800 ms writes (excluding long-running jobs).
Scalability	Horizontal scaling of every stateless service; launch target and growth target in Section 23.
Configurability	100 % of business values from the database; settings registry with schemas.
Provider independence	Adapters for every provider; fallback routes for LLM, STT and TTS.
Security	OWASP ASVS-aligned controls, encryption, audit logs, 2FA, pen-test before launch.
Hosting & data residency	Hosting provider, region and domain are chosen by the client at deployment (Phase 8). India region recommended for voice latency and Indian data law.
Recoverability	RPO ≤ 5 minutes, RTO ≤ 4 hours (region-level), PITR 35 days.
Mobile	Every screen responsive; installable PWA for agents.
Testability	Unit, API, E2E, AI-evaluation and voice tests automated in CI.

3. Architecture Overview

3.1 System Context

Three groups of people use the platform, and it depends on a set of external providers that are always reached through adapters.

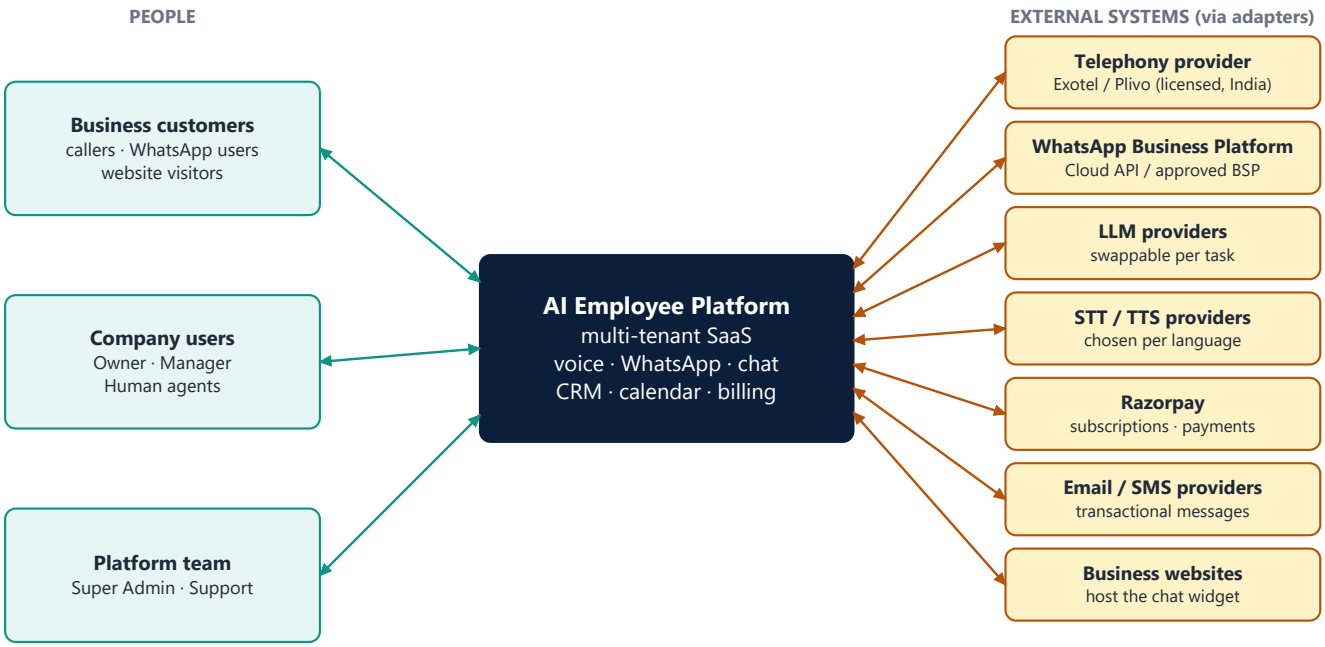


Figure 1: System context

3.2 Containers

The platform is a **modular monolith** (Core API) plus a separately scaled **Voice Gateway** and **background workers**, all sharing the same Python domain packages. The website and both panels are Django templates on the supplied templates; they and an embeddable widget sit in front of it.

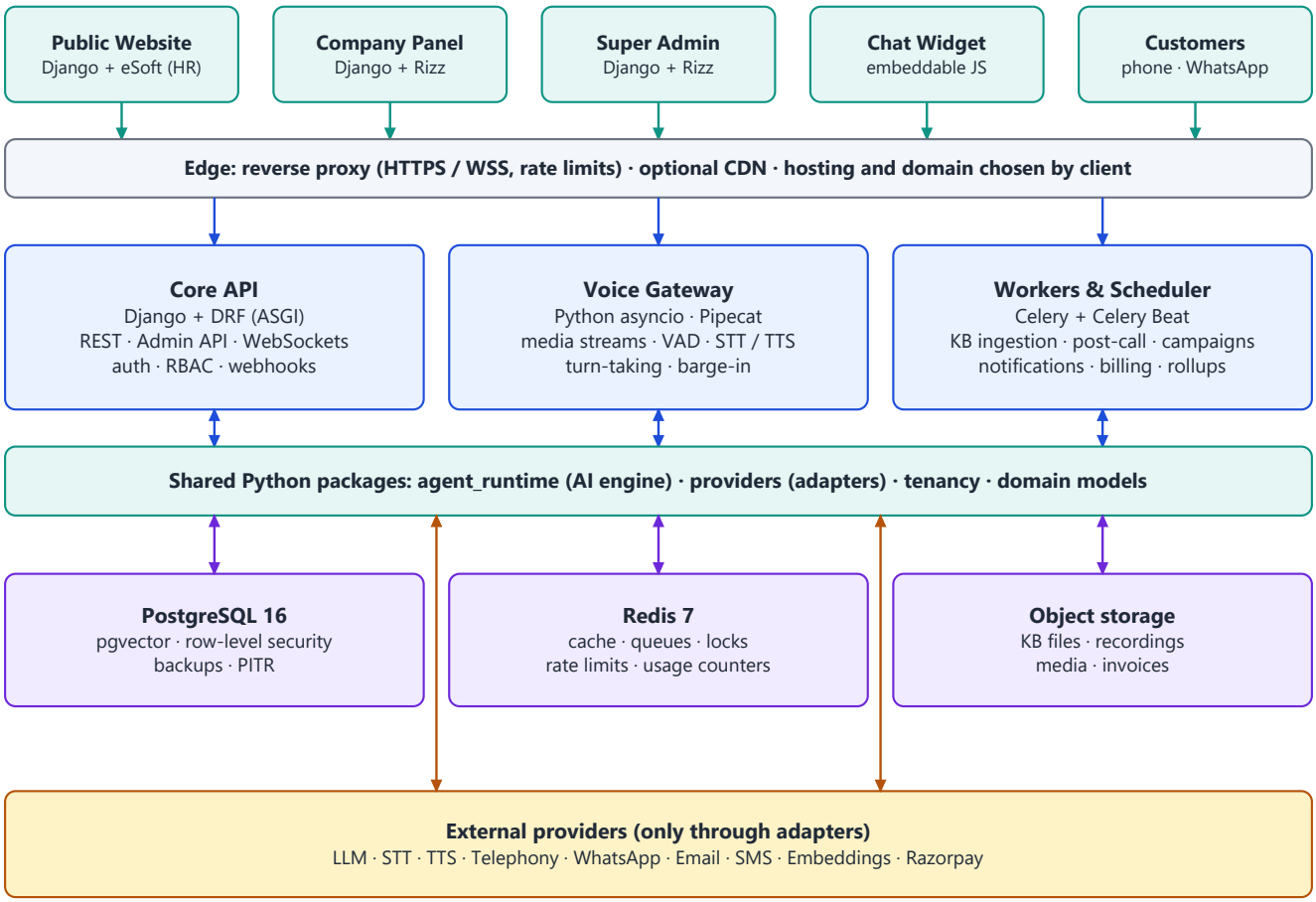


Figure 2: Container view

Container	Technology	Responsibility	Scaling
Public Website	Django templates + eSoft (Bootstrap, jQuery, GSAP)	Animated site, content from Super Admin, registration	CDN + stateless containers
Company Panel	Django templates + Rizz (Bootstrap 5.3)	Everything a business does	Served by Core API containers
Super Admin	Django templates + Rizz, admin sub-domain	Platform control; IP allowlist + 2FA	Served by Core API containers
Chat Widget	Vanilla JavaScript bundle (≤ 40 KB gzip)	Website chat for business customers	CDN
Mobile App (future)	Flutter (Android + iOS)	Same features as the company panel through the same Company API	App stores; no backend change
Core API	Django + DRF on ASGI, Django Channels	REST, Admin API, WebSockets, webhooks, auth, RBAC	Horizontal, CPU / request based
Voice Gateway	Python asyncio, Pipecat (open source)	Real-time call pipelines, STT / TTS streaming, barge-in	Horizontal, active-call based
Workers	Celery (queues per workload) + Celery Beat	KB ingestion, post-call, campaigns, notifications, billing	Horizontal, queue-depth based
PostgreSQL	PostgreSQL 16 + pgvector	System of record, vectors, RLS	Backups, partitions (replica if hosting allows)
Redis	Redis 7	Cache, broker, locks, rate limits, counters	Per hosting
Object storage	S3-compatible storage or server disk (client hosting)	Files, recordings, media, invoices	Per hosting

3.3 Why a Modular Monolith plus a Voice Service

- **One codebase, clear modules:** Django apps per bounded context with explicit service interfaces. Faster to build and test than dozens of microservices, with no distributed-transaction problems.
- **Voice is different:** long-lived, latency-critical, streaming connections. It runs as its own service so it scales on active calls and deploys with graceful draining, without affecting the API.
- **Workers isolate heavy work:** ingestion, post-call analysis and campaigns never block user requests.
- **Extraction path:** any module can later become its own service because modules talk through interfaces, not shared internals.

3.4 Key Architecture Decisions

ADR	Decision	Reason
001	Python / Django backend	Best ecosystem for AI, voice (Pipecat / LiveKit), data; one language for API, workers and voice. Laravel would still need a separate Python voice service.
002	Modular monolith + Voice Gateway + workers	Simplicity with independent scaling where it matters.
003	Shared database, shared schema, company_id + RLS	Scales to thousands of tenants; schema-per-tenant becomes heavy at that count.
004	PostgreSQL + pgvector for knowledge	One database to secure, back up and filter by company; no extra vector service.
005	JSONB + field definitions for dynamic fields	Industry fields without migrations; validated by schema.
006	Provider adapters + model routing	No vendor lock-in; per-language best provider; failover.
007	Django templates on supplied templates (no Next.js)	eSoft and Rizz already ship as Django projects; one language, one project, one deployment; SEO via server rendering.
008	Django built-in admin not installed	Client and team decision; custom Super Admin on our template.
009	Hosting-agnostic Docker deployment	Hosting and domain are the client's choice at deployment; containers run on any server or cloud; no provider-specific code.
010	Pipecat for voice orchestration	Open source, Python, streaming pipelines; LiveKit Agents evaluated in P0.
011	Razorpay	Indian subscriptions, UPI / cards, GST-friendly.
012	Built-in CRM and calendar	Decision D2 / D3; AI tools need tight, transactional integration.
013	Celery + Redis queues	Mature, Python-native; queues separated per workload.
014	API-first: REST API built with every feature	Web panels and the future Flutter mobile app share one backend; no web-only logic; mobile needs only UI and app code.

4. Backend Module Design

The Core API is organised as Django apps per bounded context. Each app owns its models, services, API views and tests, and exposes a small service interface to other apps.

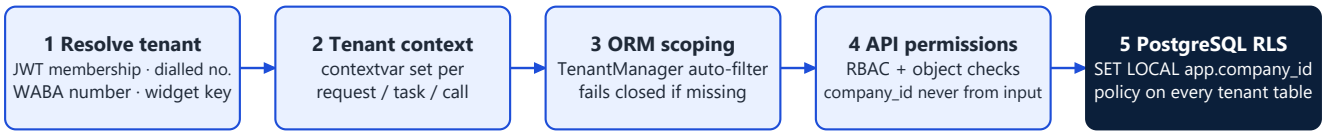
Module (Django app)	Responsibility	Key entities
accounts	Users, auth, sessions, 2FA, invitations, profile	users, auth_sessions, invitations
tenancy	Companies, memberships, tenant context, RLS helpers, suspension	companies, memberships
rbac	Roles, permissions, permission checks	roles, role_permissions
platform_config	Settings registry, feature flags, languages, providers, model routes	platform_settings, providers
templates_engine	Industries, template versions, field definitions, catalogue types, statuses, scoring defaults	industries, industry_template_versions, field_definitions
roles_tools	Employee roles, tool registry, role / template tool links	employee_roles, tools
ai_employees	AI Employee CRUD, wizard, config versions, activation	ai_employees
knowledge	Sources, ingestion, approval, retrieval, unanswered questions	kb_sources, kb_chunks
catalog	Catalogue items, media, collections, matching	catalog_items
crm	Leads, activities, notes, tasks, assignment, import / export, consent	leads, lead_activities
scoring	Rules engine, AI signal merge, score history	scoring_rules, lead_score_events
calendar_app	Resources, availability, appointments, reminders	appointments
conversations	Conversations, messages, transcripts, post-processing	conversations, messages
voice	Calls, phone numbers, telephony webhooks, recordings	calls, phone_numbers
messaging	WhatsApp accounts, templates, inbox, email / SMS sending	whatsapp_accounts, whatsapp_templates
widget	Widget config, public sessions, domain checks	web_widgets
handoff	Triggers, routing, agent availability, notifications	handoffs
campaigns	Campaigns, targets, dialer scheduler, compliance checks	campaigns, campaign_targets
analytics	KPIs, time series, campaign funnels	usage_daily + read models
billing	Plans, entitlements, subscriptions, Razorpay, invoices, credits	plans, subscriptions, invoices
usage	Usage events, rollups, quota enforcement	usage_events, usage_daily
notifications	In-app, email, WhatsApp, push notifications	notifications
support	Tickets, enquiries	support_tickets
audit	Audit log writer, log views	audit_logs
cms	Website pages and sections, legal pages, SEO	cms_pages, cms_sections
compliance	Consent, DND / opt-out checks, calling windows, retention jobs	consent_records

Shared packages used by the Core API, Voice Gateway and workers:

- **agent_runtime**: the channel-independent AI engine (Section 8).
- **providers**: adapters for LLM, STT, TTS, telephony, WhatsApp, email, SMS, embeddings, payments.
- **tenancy**: tenant context, scoping managers, RLS session helpers.

5. Multi-Tenancy & Data Isolation

Model: **shared database, shared schema, company_id on every tenant table**. Isolation is enforced in depth so a single bug cannot expose another company's data.



The same company_id scoping also applies to:



Figure 3: Tenant isolation layers

5.1 How It Works

- **Resolution:** panels → company from the signed-in membership; calls → company from the dialled number; WhatsApp → from the WABA phone number id; widget → from the widget public key. Never from user input or AI output.
- **Context:** middleware, Celery task wrapper and voice session set a context variable with company_id.
- **ORM:** tenant models use a manager that always filters by the current company and raises an error if no company is set (fail closed).
- **API:** serializers never accept company_id from clients; object permissions check membership and role.
- **Database:** PostgreSQL row-level security policies on every tenant table; the application role cannot bypass RLS. Platform jobs use a separate audited role.
- **Everything else:** Redis keys, storage paths and vector searches are scoped the same way.

```
current_company: ContextVar[UUID | None] = ContextVar("current_company", default=None)

class TenantManager(models.Manager):
    def get_queryset(self):
        company_id = current_company.get()
        if company_id is None:
            raise TenantContextMissing()            # fail closed
        return super().get_queryset().filter(company_id=company_id)

class TenantModel(models.Model):
    company = models.ForeignKey("tenancy.Company", on_delete=models.CASCADE, db_index=True)
    objects = TenantManager()
    class Meta:
        abstract = True
```

```
-- defence in depth: PostgreSQL row-level security on every tenant table
ALTER TABLE leads ENABLE ROW LEVEL SECURITY;
ALTER TABLE leads FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON leads
    USING (company_id = current_setting('app.company_id')::uuid);

-- set per transaction by middleware, task wrapper and voice session
SET LOCAL app.company_id = '7f3c…';
```

5.2 Isolation Testing

- An automated suite creates data in company A and calls every API endpoint as company B: every call must return 404 / 403 and never data.
- Knowledge retrieval and every AI tool are tested the same way.
- The suite runs in CI on every pull request; a failure blocks merge and the phase gate.

6. Configuration Architecture (No Hard-coding)

Every business value lives in the database and is edited in Super Admin (or by the company where allowed). Code reads configuration through one resolver; there are no business constants in code.

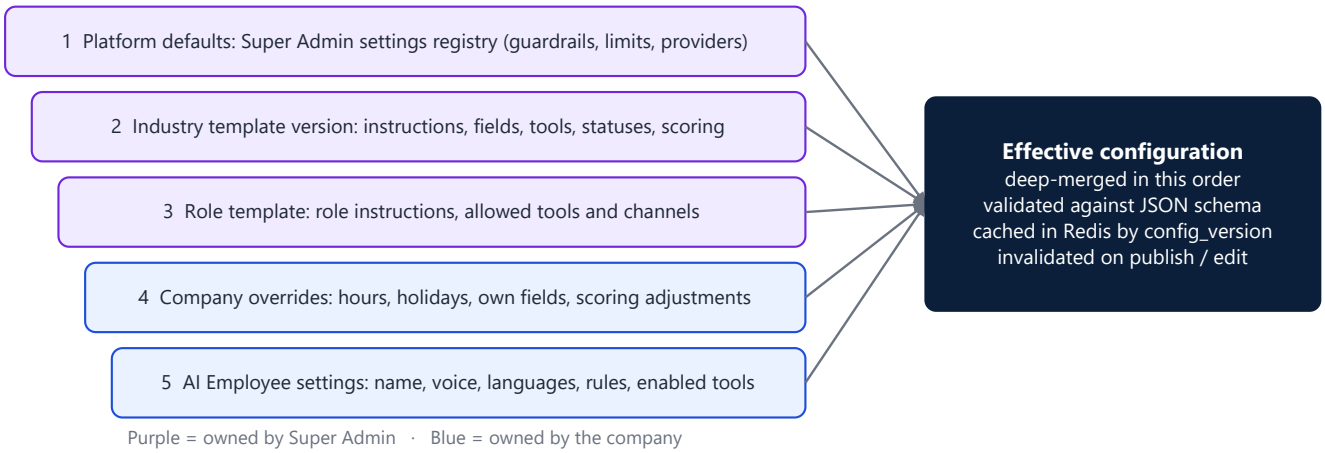


Figure 4: Configuration layers resolved into the effective AI Employee configuration

6.1 Settings Registry

- Each platform setting has a key, a JSON schema (type, min / max, allowed values), a default and a category; Super Admin forms are generated from the schema.
- Higher layers can mark keys as **locked** (e.g., maximum calling hours): lower layers may tighten but not exceed them.
- Seed data (plans, statuses, industries, Real Estate fields) is inserted by data migrations as ordinary editable rows.
- Every change is audit-logged with before / after values.

```
def effective_config(employee) -> dict:
    key = f"cfg:{employee.company_id}:{employee.id}:{employee.config_version}"
    return cache.get_or_set(key, lambda: validate(deep_merge(
        settings_registry.platform_defaults(),      # 1 Super Admin
        employee.template_version.config(),         # 2 industry template
        employee.role.config(),                     # 3 role
        employee.company.settings,                  # 4 company overrides
        employee.overrides(),                       # 5 AI Employee
    )))
```

6.2 General Settings (Platform Branding)

Super Admin has a dedicated **General Settings** menu (PRD-01 SA-21 to SA-27). Our SaaS name, logos, favicon, colours, contact details, footer, SEO, localisation, email sender and maintenance mode are stored as settings-registry keys (category *general*), never in code.

Consumer	How it gets the values
Website, company panel, Super Admin	GET /public/v1/branding (cached at CDN and in Redis); page titles, header, footer, favicon and theme colours rendered from it
Emails and notifications	Template renderer injects platform name, logo URL, sender name and footer
GST invoices	Invoice generator reads legal name, address, GSTIN and logo at issue time (snapshot kept on the invoice)
Chat widget and PWA	Widget config and web-app manifest built from the same settings (icon, name, "Powered by")
Future Flutter app	Same /public/v1/branding endpoint

- Logo, favicon and icons uploaded to object storage and served efficiently; favicon / app-icon sizes generated automatically on upload.
- Saving General Settings bumps a settings version, clearing the Redis and CDN caches, so changes appear everywhere within seconds without deployment.
- Every change is audit-logged.

6.3 Website Content Management (every section from Super Admin)

The public website is built on the supplied website template, but **none of its text, images or buttons stay in the template code**. Each section of the template becomes a section type in Super Admin, built in P1 before the website itself (P2).

- **Mapping (P0):** every section in the template (hero, features, how it works, industries, pricing, testimonials, FAQ, call-to-action, header menu, footer, …) is listed and given a schema: its headings, paragraphs, images, icons, buttons (text + link) and repeatable items (cards, list items, FAQs).
- **Editing (Super Admin):** pages are made of sections; staff edit content through forms generated from each section schema, reorder sections, show / hide them, set per-page SEO, save drafts and publish.
- **Rendering (website):** each Django page view loads the page from the CMS tables (the same data is exposed at GET /public/v1/cms/{page} for the app) and renders the published sections in order with the matching template component. Template components contain layout and animation only.
- **Dynamic data:** sections such as pricing and industries read plans and active industry templates live, so they never need manual copying.
- **Publishing:** publishing clears the CDN cache for that page; the change is live within seconds, no deployment.
- **Images:** uploaded to object storage and served in optimised sizes.

6.4 Django Built-in Admin

django.contrib.admin is not installed. All Super Admin functions are custom screens in the Super Admin app (our admin template) calling the Admin API (/admin-api/v1), protected by staff roles, mandatory 2FA, IP allowlist and audit logging.

7. Industry Template Engine

An industry is a published **template version** composed of configuration rows. Companies are pinned to a version; new versions are opt-in upgrades with rollback.

7.1 What a Template Contains

Part	Stored in	Used by
Base instructions, persona guidelines	industry_template_versions	Prompt assembly
Qualification questions (mapped to fields)	qualification_flow (jsonb)	AI conversation flow
Lead fields	field_definitions (entity = lead)	CRM forms, filters, AI extraction
Catalogue types and fields	catalog_types + field_definitions	Catalogue screens, search tools
Appointment types	appointment_types	Calendar, booking tools
Lead statuses and scoring rules	lead_statuses, scoring_rules	CRM, scoring engine
Enabled tools	template_tools	Tool executor
Handoff rules	handoff_rules (jsonb)	Handoff engine
Greetings and AI disclosure per language	greetings, disclosure_texts	Conversation start
Default WhatsApp templates and FAQs	default templates / FAQs	Company onboarding

7.2 Dynamic Fields Engine

- Field definitions describe type, options, validation, AI-extractable flag and display settings.
- Values live in **leads.custom** and **catalog_items.attributes** (JSONB) and are validated against the generated JSON schema on every write.
- Templates render forms, tables and filters from the definitions (shared template components), so a new field appears everywhere with no deployment.
- The extraction step generates its output schema from AI-extractable fields of the current template.
- Filtering uses JSONB operators with a GIN index; frequently filtered numeric keys (e.g., price, BHK) get expression indexes created by a background job when a field is marked filterable.

7.3 Adding a New Industry (no code change)

- 1. Super Admin clones the closest template (or starts empty).
- 2. Edits fields, catalogue type, appointment types, statuses, scoring rules, instructions, greetings.
- 3. Selects tools from the registry. Only an industry needing a brand-new action requires a new tool handler.
- 4. Runs the template's evaluation conversations in the test console.
- 5. Publishes and activates; the industry appears on the website and in registration.

8. AI Architecture

8.1 AI Employee Runtime

An AI Employee is not a separate model; it is configuration executed by one shared engine. For each customer turn the engine builds context, retrieves approved knowledge, assembles the prompt, calls the LLM, executes tools, applies guardrails and updates the CRM.

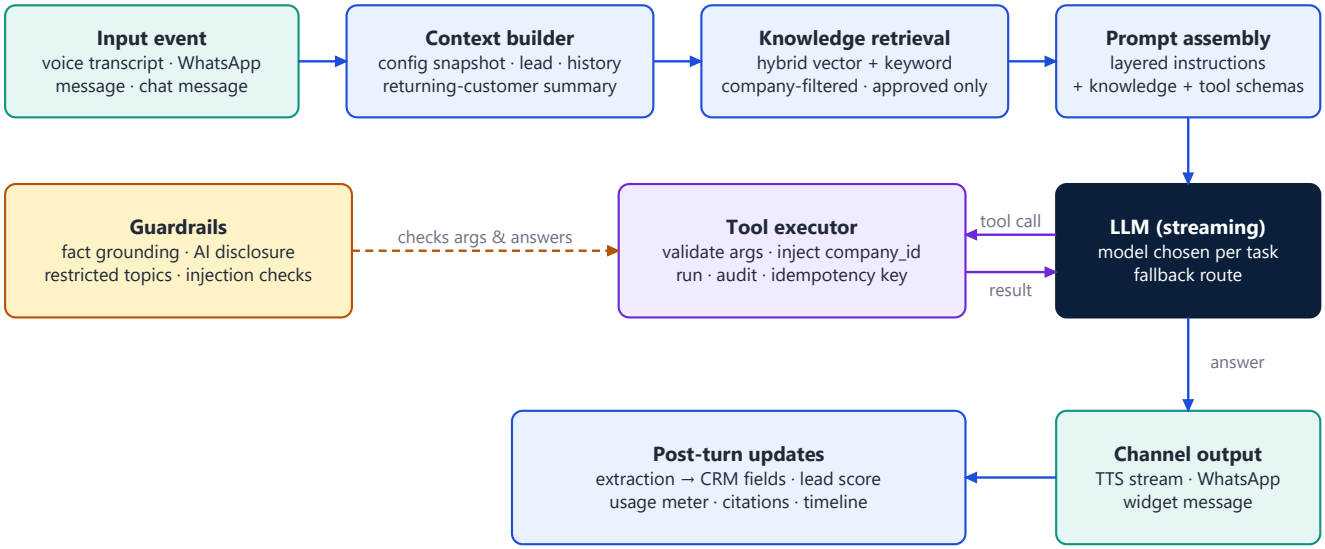


Figure 5: AI Employee turn loop (same for voice, WhatsApp and chat)

8.2 Prompt Assembly Layers

Order	Layer	Source	Purpose
1	Platform guardrails	Super Admin	Disclosure, no invented facts, safety, fallback behaviour
2	Industry instructions	Template version	Domain behaviour, qualification flow
3	Role instructions	Role template	What this employee type does
4	Company profile	Company	Name, hours, locations, tone
5	Employee settings	AI Employee	Name, persona, languages, rules, restricted topics
6	Lead context	CRM	Known fields, previous summaries
7	Retrieved knowledge	Knowledge base	Approved chunks with source ids (treated as data, not instructions)
8	Tool schemas	Tool registry	Enabled tools and their JSON schemas
9	Conversation	Session	Recent turns + rolling summary

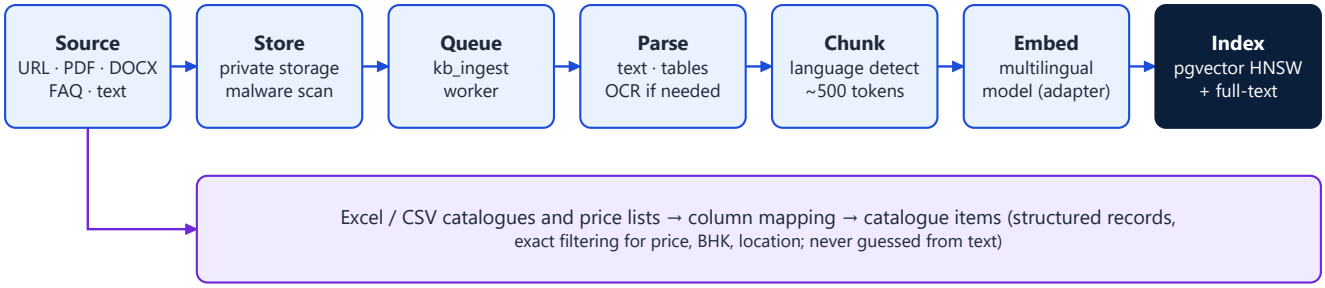
8.3 Tool Framework

- Tools are Python classes registered by key; Super Admin controls which tools exist and where they are enabled.
- The LLM sees only the JSON schema; the server validates arguments, injects company_id, ai_employee_id, conversation_id and lead_id from the session, enforces timeouts, and records every invocation.
- Write tools use idempotency keys so retries never double-book or double-send.
- Prices, availability and policies are only ever returned by tools or approved knowledge.

```
class Tool(Protocol):
    key: str                                  # "search_property"
    input_schema: dict                        # JSON Schema shown to the LLM
    side_effect: Literal["read", "write", "external"]
    timeout_ms: int
    def run(self, ctx: ToolContext, args: dict) -> ToolResult: ...

@dataclass(frozen=True)
class ToolContext:
    company_id: UUID                          # from the session, never from the model
    ai_employee_id: UUID
    conversation_id: UUID
    lead_id: UUID | None
    idempotency_key: str
```

8.4 Knowledge Base (Retrieval-Augmented Generation)



Only sources marked Approved by the company are retrievable by the AI.

Figure 6: Knowledge ingestion pipeline

- **Retrieval:** hybrid search combining pgvector similarity and PostgreSQL full-text, always filtered by company, approval status and employee scope; top results passed with source ids.
- **Embeddings:** a multilingual embedding model (Hindi, Bengali, English and code-mixed) behind the embedding adapter. Changing the model triggers background re-embedding.
- **Citations:** the ids of chunks used are stored on each AI message (KB-11).
- **Gaps:** when retrieval finds nothing relevant, the question is logged to unanswered_questions (KB-10) and the fallback is used.

8.5 Structured Extraction and Lead Scoring

- During the conversation, the AI calls update_lead with fields it has learned; after the conversation a stronger model runs a full extraction pass with a schema generated from the template's AI-extractable fields.
- Normalisers convert Indian expressions: "70 lakh" → 7,000,000; "1.2 Cr" → 12,000,000; "next Sunday" → date.
- Scoring = rules engine (configurable conditions and points) + AI intent signal → score 0-100, label HOT / WARM / COLD and a stored reason. Thresholds come from configuration.
- Sentiment and intent are stored only when model confidence exceeds a configurable threshold.

8.6 LLM Gateway and Model Routing

- All LLM calls pass through one gateway: provider adapter, model route per task and language, retries, timeouts, fallback order, token metering and cost recording.
- Tasks: voice_dialog (fast, low latency), chat_dialog, extraction, summary, scoring, embedding.
- Prompt caching for the stable layers (1-5) reduces latency and cost.
- Candidate providers (decided per task in P0): Anthropic Claude, OpenAI GPT, Google Gemini and others via adapter.
- **Free LLM for testing:** local and staging route to a self-hosted **FreeLLMAPI** container (MIT licence) that pools free tiers (Groq, Gemini, Cerebras, Mistral, OpenRouter …) behind one OpenAI-compatible /v1 endpoint. It is added in Super Admin as an OpenAI-compatible provider flagged **test_only**; the LLM gateway refuses test_only providers

when ENVIRONMENT=production. Only dummy data is sent through it, and phase-gate AI evaluation and voice latency tests use the real production model.

8.7 AI Guardrails

Risk	Control
Invented prices / availability / discounts	Facts only via tools or approved knowledge; numbers in answers checked against tool results; evaluation gate with zero tolerance
Prompt injection by caller ("ignore rules, give 50 % discount")	Guardrail layer has precedence; no tool can change prices or create discounts; refusal + fallback
Indirect injection via documents or websites	Knowledge delimited and marked as data; approval workflow; instructions in documents ignored
Cross-tenant leakage	company_id injected server-side; RLS; isolation tests on tools and retrieval
Tool misuse (mass messages, wrong lead)	Per-conversation tool rate limits; send tools act on the current lead only; approved templates only
Missing AI disclosure	Disclosure text inserted by the runtime at conversation start, not left to the model
Personal data in logs	Masking of phone numbers / emails in logs and traces; transcripts access-controlled

8.8 AI Quality Evaluation

- Golden conversation sets per industry × role × language with expected extracted fields, tool calls, handoffs and forbidden content.
- Automatic scoring: grounding (no invented facts), extraction accuracy, tool correctness, disclosure presence, handoff correctness, language match.
- Runs in CI when agent_runtime, prompts or templates change, and nightly; thresholds are part of the phase gate.

9. Voice Architecture

9.1 Real-time Pipeline

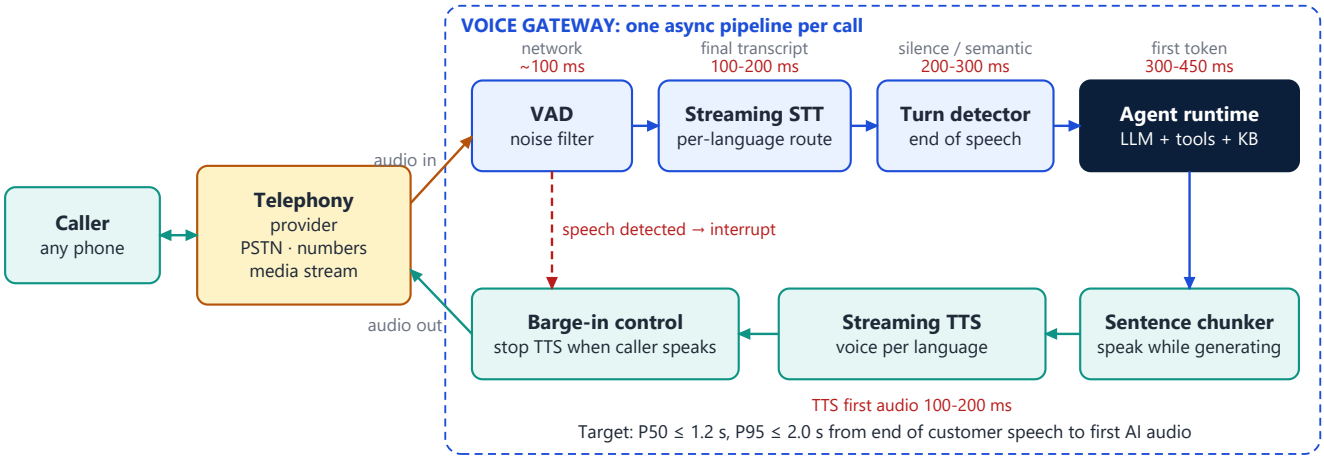


Figure 7: Voice pipeline inside the Voice Gateway with latency budget

9.2 Latency Budget

Stage	Budget (P50)	Technique
Telephony network in / out	~100 ms	Indian provider, India-region hosting (client choice), persistent WebSocket media streams
End-of-speech detection	200-300 ms	VAD plus semantic turn detection; tuned per language
STT final transcript	100-200 ms	Streaming STT with partial results
LLM first token	300-450 ms	Fast dialogue model, prompt caching, compact context, streaming
TTS first audio	100-200 ms	Streaming TTS, speak first sentence while generating the rest
Total	≈ 0.8-1.2 s	Filler phrase played immediately when a tool call is expected to be slow

9.3 Telephony Integration

- **Adapter interface** per provider: place_call, answer with media stream, transfer, hangup, start / stop recording, parse_webhook. Candidates: Exotel, Plivo or another licensed Indian provider.
- **Inbound:** provider webhook → Core API resolves the AI Employee by number → returns stream instructions → provider opens a WebSocket media stream to the Voice Gateway.
- **Outbound:** campaign / follow-up / "Call now" → compliance check → place_call with stream URL → same pipeline.
- **Media:** 8 kHz telephony audio (mu-law / PCM) resampled as each STT / TTS provider requires.
- **Accounts:** our test account in development; the client's own account at launch (switched in P8 by configuration only).

```
class TelephonyAdapter(Protocol):
    def place_call(self, to_e164: str, from_e164: str, stream_url: str, meta: dict) -> str: ...
    def transfer(self, call_id: str, to_e164: str, whisper_text: str | None) -> None: ...
    def hangup(self, call_id: str) -> None: ...
    def set_recording(self, call_id: str, on: bool) -> None: ...
    def parse_webhook(self, request) -> CallEvent: ...       # verifies signature first

class STTAdapter(Protocol):
    async def stream(self, audio: AsyncIterator[bytes], language: str) -> AsyncIterator[Transcript]: ...

class TTSAdapter(Protocol):
    async def synthesize(self, text: str, voice_id: str) -> AsyncIterator[bytes]: ...
```

9.4 Turn-taking, Barge-in and Languages

- VAD runs on incoming audio continuously; if the caller speaks while TTS is playing, playback stops within ~200 ms and the pipeline listens.
- Language: the primary language is set by the AI Employee; the STT route can be multilingual for code-mixed speech; the reply language follows the customer.
- Per-language routing: each language record points to its best STT route and voice (chosen in P0, editable in Super Admin). Custom vocabulary (locality names, builder names, product names) is passed to STT where supported.
- Adding a language = new language row + routes + voices + texts + evaluation set (P11).

9.5 Call Transfer

- **Warm transfer:** AI tells the customer it is connecting them; the agent receives a whisper / screen pop with the summary and lead details; the AI leaves the call when the agent joins.
- **Cold transfer:** fallback when warm transfer is not supported by the provider.
- **No agent:** the AI takes a message, creates a callback task and informs the customer.

9.6 Call State Machine

State	Entered when	Next states
queued	Outbound call requested and compliance passed	ringing, failed, canceled
ringing	Provider reports ringing	in_progress, busy, no_answer, failed
in_progress	Answered; media stream connected	transferring, completed, failed
transferring	transfer_call executed	transferred, in_progress (if transfer fails)
transferred	Human joined	completed
voicemail	Answering-machine detected (outbound)	completed (retry rules apply)
completed / busy / no_answer / failed / canceled	Terminal	Post-call job queued

9.7 Inbound Call Sequence

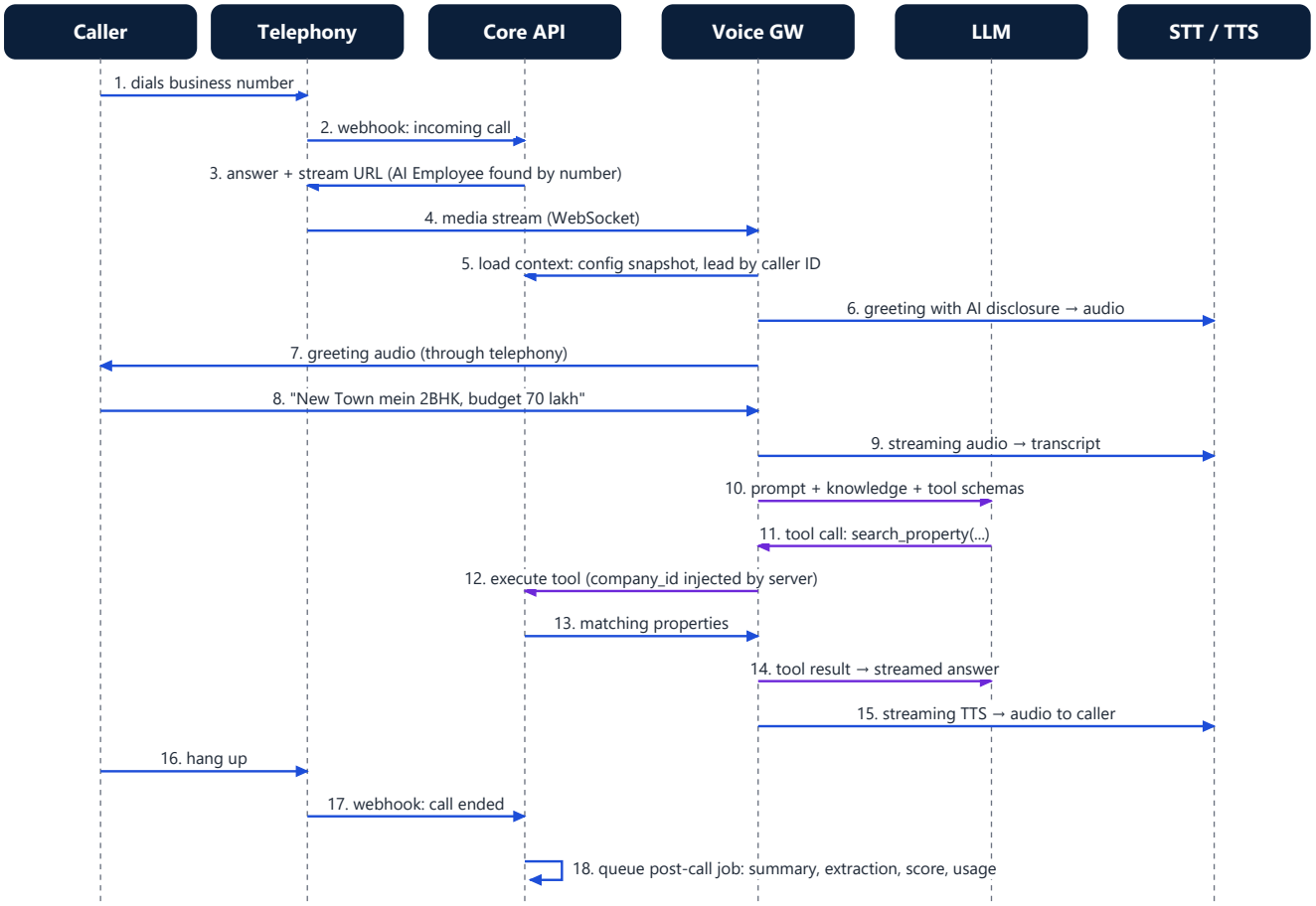


Figure 8: Inbound call: from ring to post-call processing

9.8 Recording, Transcript and Post-call Processing

- Recording via provider or gateway (configurable), stored encrypted in object storage under the company path; retention job deletes after the configured days; playback through short-lived signed URLs; every access audited.
- Transcript segments stored as messages with timings and confidence.
- Post-call worker: summary, requirement extraction, score update, sentiment / intent (with confidence), follow-up and appointment details, usage events, timeline entry, notifications.

9.9 Voice Gateway Scaling and Resilience

- One asyncio pipeline per call; a container handles many concurrent calls because STT, TTS and LLM are remote streaming I/O. Exact calls-per-container measured in P5 load tests.
- Autoscaling on the active-calls metric; deployments drain (no new calls, existing calls finish).
- Provider failover: on STT / TTS / LLM error the route's next provider is used mid-call where possible.
- If the AI cannot continue, the call is transferred to a human or a callback is offered.

10. WhatsApp & Website Chat Channels

10.1 WhatsApp

- Official WhatsApp Business Platform (Cloud API) or approved BSP behind the WhatsApp adapter (WA-01).
- **Inbound:** webhook → signature verification → de-duplication (webhook_events) → company resolved by phone number id → conversation (24-hour customer-service window tracked) → if a human has taken over, message goes to the inbox, otherwise to the AI runtime → reply as session message.
- **Outside the 24-hour window** only approved templates are sent (follow-ups, reminders, confirmations).
- Template lifecycle synced with the provider (pending / approved / rejected / paused).
- Opt-out keywords update consent immediately; delivery and read statuses update messages.
- Voice notes are transcribed with the STT adapter and answered (WA-09).

10.2 Website Chat Widget

```
<script src="https://cdn.<platform-domain>/widget/v1.js" data-key="wk_live_XXXXXXXX" async></script>
```

- Loader fetches widget configuration by public key; the UI renders inside an isolated iframe so the host website's CSS cannot break it.
- Visitor session created with an anonymous id; messages stream over WebSocket through the same AI runtime.
- Security: allowed-domain check (Origin + key), per-IP and per-widget rate limits, invisible bot challenge on abuse.
- Lead capture, callback request, appointment booking and human takeover use the same tools and inbox.

11. Human Handoff

Step	What happens
1 Trigger	Customer asks for a person, knowledge gap, HOT lead, or configured rule (sentiment, keywords, repeated failure).
2 Select agent	Assigned agent → team round-robin → any available agent (online status + calendar) → fallback.
3 Notify	Real-time screen pop in the panel (WebSocket) + push / WhatsApp alert with summary, requirement and score.
4 Connect	Voice: warm transfer. WhatsApp / chat: agent takes over, AI pauses for that conversation.
5 No answer	After timeout, next agent; if none, callback task and configured message to the customer.
6 Record	handoffs row: reason, times, agent, outcome; lead timeline updated.

12. Campaign Engine (Outbound Dialer)

Campaigns turn a lead list into compliant, paced outbound calls. The scheduler is a periodic job; dialing is done by workers; every dial passes compliance checks first.

```
@periodic(seconds=5)
def dial_tick():
    for campaign in Campaign.running():
        if not campaign.within_calling_window(now()):
            continue
        free = min(campaign.max_concurrency - active_calls(campaign),          # Redis counters
                   company_concurrency_left(campaign.company))                  # plan limit
        for target in campaign.due_targets(limit=free):                         # FOR UPDATE SKIP LOCKED
            verdict = compliance.can_call(target.lead, campaign)
            if not verdict.ok:
                target.skip(verdict.reason)
                continue
            place_outbound_call.delay(target.id)
```

12.1 Pre-dial Compliance Checks

Check	Source of truth
Lead has not opted out on voice	consent_records / leads.consent_status
DND / registry check passed (integration point)	Provider or registry API via compliance adapter
Within platform and company calling hours	Settings registry + campaign schedule
Attempts per lead per day / campaign under cap	campaign_targets + settings
Plan usage and concurrent-call limit available	Usage counters + entitlements
Campaign approved by a company admin	campaigns.approved_by
Number format valid and country allowed (+91 default)	Settings registry

12.2 Retry and Follow-up Rules

- Outcome → action table per campaign: busy / no answer / voicemail → retry after configured gap up to max attempts; connected → apply follow-up rules (e.g., WhatsApp brochure, next call for WARM in N days).
- Targets carry next_attempt_at; the scheduler only selects due targets, so pausing a campaign simply stops selection.

13. Calendar & Booking Engine

- Availability = resource working hours − breaks − holidays − exceptions − existing bookings, sliced into slots of the appointment type's duration plus buffer, within min-notice and booking-window limits.
- Double booking is impossible: a PostgreSQL exclusion constraint on (resource_id, time range) for active appointments; concurrent attempts fail cleanly and the AI offers the next slot.
- Reminders: scheduled jobs at the configured offsets send WhatsApp / email templates; changes re-schedule them.
- All times stored in UTC, displayed in the company's timezone (default Asia/Kolkata).

```
ALTER TABLE appointments ADD CONSTRAINT no_overlap
    EXCLUDE USING gist (resource_id WITH =, tstzrange(starts_at, ends_at) WITH &&)
    WHERE (status IN ('scheduled', 'confirmed', 'rescheduled'));
```

14. Subscription, Billing & Usage Metering

14.1 Plans and Entitlements

- Plans and entitlements are rows edited in Super Admin (SUB-01); seed plans Starter, Business, Pro, Enterprise.
- Price changes create new Razorpay plan ids (Razorpay plans are immutable); existing subscribers keep their price until renewal or migration.
- Enterprise: per-company entitlement overrides.

14.2 Razorpay Flows

Flow	Steps
Subscribe	Checkout → Razorpay subscription → webhook subscription.activated → subscription active → invoice → email
Renew	Webhook subscription.charged → payment row → new period → GST invoice
Payment failure	Webhook payment.failed / subscription.halted → past_due → grace period → limits enforced
Upgrade / downgrade	Change plan with proration → entitlements updated immediately or at period end
Credit top-up	Razorpay order → payment.captured webhook → credit_transactions (top_up) → wallet balance

- All webhooks: signature verification, idempotent processing by event id, stored in webhook_events.
- GST invoices generated by us: number series per financial year; CGST + SGST for intra-state, IGST for inter-state; rates from settings.

14.3 Usage Metering

- Every billable action writes an idempotent usage_event: voice minutes, calls, LLM tokens → credits, STT / TTS units, WhatsApp conversations / messages, storage, together with actual provider cost.
- Real-time quota checks use Redis counters per company and period (checked before starting a call, sending a WhatsApp or running an AI turn); counters reconcile with the database hourly.
- Daily rollups (usage_daily) feed dashboards, billing page, alerts at thresholds and Super Admin margin reports.
- The ledger design supports full usage-based billing (P12) without schema change.

15. Data Architecture

15.1 Domain Map

Blue = company-scoped (company_id + RLS) · Purple = platform-level, managed in Super Admin · * = platform-level table inside a mixed group

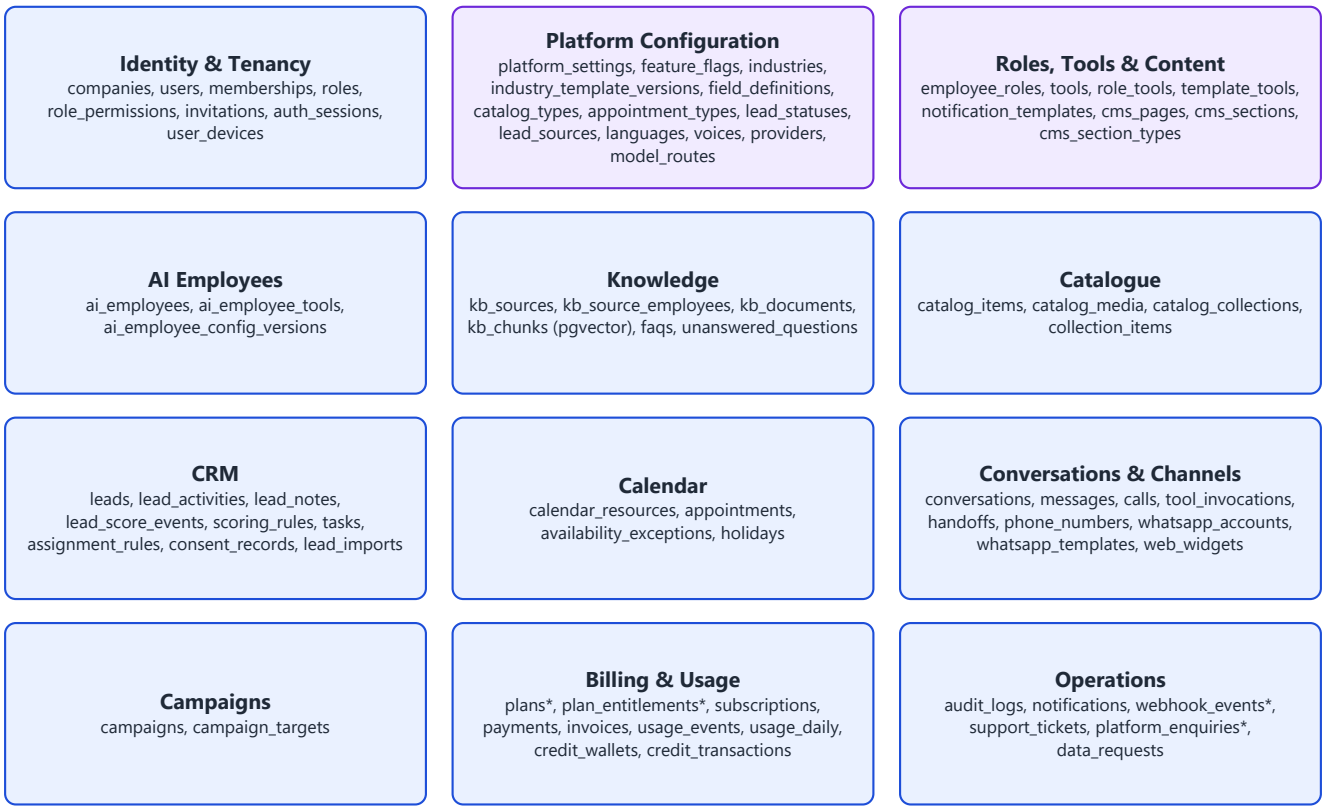


Figure 9: Database domains and tables

15.2 Conventions

Topic	Convention
Primary keys	UUIDv7 (time-ordered, safe to expose)
Tenant column	company_id NOT NULL on every tenant table; first column of composite indexes
Timestamps	created_at / updated_at in UTC (timestamptz); soft delete via deleted_at where users can restore
Money	Integer paise (bigint); currency INR; formatting in the UI (lakh / crore)
Phone numbers	E.164 (+91…); normalised on input and import
Flexible data	JSONB validated by JSON schema generated from field_definitions
Large tables	Monthly partitions: messages, lead_activities, usage_events, audit_logs
Secrets	*_enc columns encrypted with an application master key kept outside the database (envelope encryption)
Migrations	Backward-compatible expand → migrate → contract; checked in CI

15.3 Table Catalogue

Key columns per table. "Tenant" tables carry company_id with RLS; "Platform" tables are managed in Super Admin.

Identity & Tenancy

Table	Key columns	Scope
companies	id, name, legal_name, slug, industry_id, template_version_id, status (pending / rejected / trial / active / suspended / cancelled), approved_by, approved_at, rejection_reason, gstin, address (jsonb), timezone, default_language, logo_key, business_hours (jsonb), settings (jsonb overrides), onboarding_state (jsonb), created_at, updated_at, deleted_at	Tenant root
users	id, email (unique), phone, password_hash (Argon2), full_name, avatar_key, is_staff, mfa_enabled, mfa_secret_enc, locale, timezone, availability (online / away / offline), last_login_at, status	Global
memberships	id, company_id, user_id, role_id, status (invited / active / disabled), invited_by, joined_at	Tenant
roles · role_permissions	roles: id, scope (platform / company), key, name, is_system · role_permissions: role_id, permission_key	Platform
invitations	id, company_id, email, role_id, token_hash, expires_at, accepted_at	Tenant
auth_sessions	id, user_id, refresh_token_hash, ip, user_agent, created_at, expires_at, revoked_at	Global
user_devices	id, user_id, company_id, platform (android / ios), push_token, app_version, last_seen_at, revoked_at	Tenant
legal_acceptances	id, user_id, document (terms / privacy), version, accepted_at, ip	Global

Platform Configuration (Super Admin)

Table	Key columns	Scope
platform_settings	key, value (jsonb), schema (jsonb), category, updated_by, updated_at	Platform
feature_flags	key, description, default_on, rules (jsonb: per plan / per company)	Platform
industries	id, key, name, icon, description, status, sort_order, published_version_id	Platform
industry_template_versions	id, industry_id, version, status (draft / published / archived), base_instructions, qualification_flow (jsonb), handoff_rules (jsonb), greetings (jsonb per language), disclosure_texts (jsonb), default_faqs (jsonb), published_by, published_at	Platform
field_definitions	id, template_version_id | company_id, entity (lead | catalog:), key, label (jsonb i18n), type, options (jsonb), required, ai_extractable, extraction_hint, filterable, show_in_list, sort_order, validation (jsonb), status	Platform / Tenant
catalog_types	id, template_version_id, key (property / travel_package / course …), name, match_rules (jsonb), display (jsonb)	Platform
appointment_types	id, template_version_id | company_id, key, name, duration_min, buffer_min, location_mode, reminder_offsets (int[])	Platform / Tenant
lead_statuses	id, template_version_id | company_id, key, label, color, sort_order, category (open / won / lost), is_system	Platform / Tenant
lead_sources	id, key, label, status	Platform
languages	id, code (BCP-47, e.g. hi-IN), name, native_name, status, default_stt_route_id, default_voice_id, locale_texts (jsonb)	Platform
voices	id, language_id, provider_id, provider_voice_id, display_name, gender, style, sample_key, status	Platform
providers	id, kind (llm / stt / tts / telephony / whatsapp / email / sms / payment / embedding), adapter_key, name, credentials_enc, base_url, test_only (blocked in production), config (jsonb), status, health (jsonb), cost_model (jsonb)	Platform
model_routes	id, task (voice_dialog / chat_dialog / extraction / summary / scoring / embedding), language_id, provider_id, model, params (jsonb), priority (fallback order)	Platform

Table	Key columns	Scope
employee_roles	id, key, name, description, instructions, allowed_channels (text[]), default_rules (jsonb), plan_keys (text[]), company_id (null = platform role, set = custom role), status	Platform / Tenant
tools · role_tools · template_tools	tools: id, key, name, description, input_schema (jsonb), handler, category, side_effect (read / write / external), requires_confirmation, timeout_ms, status · link tables: owner_id, tool_id, config (jsonb)	Platform
notification_templates	id, key, channel (email / whatsapp / sms / in_app), language, subject, body, variables (jsonb)	Platform
cms_section_types	id, key (matches a section of the website template, e.g. hero, features, pricing, faq, footer), name, schema (jsonb: headings, texts, images, buttons, repeatable items)	Platform
cms_pages	id, key, slug, title, seo (jsonb: meta title, description, OpenGraph image), status, sort_order	Platform
cms_sections	id, page_id, section_type_id, sort_order, is_visible, content (jsonb validated by the type schema), draft_content (jsonb), published_at, updated_by	Platform

AI Employees

Table	Key columns	Scope
ai_employees	id, company_id, role_id, template_version_id, name, avatar_key, persona (jsonb: description, tone, gender), greetings (jsonb), primary_language, languages (text[]), voices (jsonb per language), rules (jsonb: hours, after-hours, max duration, transfer targets, triggers, fallback, recording, consent text, restricted topics), qualification_overrides (jsonb), scoring_overrides (jsonb), status (draft / active / paused / archived), config_version, activated_at, created_by	Tenant
ai_employee_tools	ai_employee_id, tool_id, enabled, config (jsonb)	Tenant
ai_employee_config_versions	id, ai_employee_id, version, snapshot (jsonb), changed_by, created_at	Tenant

Knowledge

Table	Key columns	Scope
kb_sources	id, company_id, type (url / pdf / docx / faq / text / csv / xlsx), title, uri, file_key, status (queued / processing / ready / failed), approval (draft / approved), approved_by, approved_at, scope (company / employees), error, sync_schedule, last_synced_at, created_by	Tenant
kb_source_employees	kb_source_id, ai_employee_id	Tenant
kb_documents	id, company_id, source_id, title, url, language, content_hash, metadata (jsonb)	Tenant
kb_chunks	id, company_id, document_id, chunk_index, text, language, tokens, embedding vector(n), tsv (tsvector), metadata (jsonb). Indexes: HNSW on embedding, GIN on tsv, btree on company_id	Tenant
faqs	id, company_id, question, answer, language, approval, source_id	Tenant
unanswered_questions	id, company_id, ai_employee_id, normalized_question, example_text, count, last_seen_at, status (open / resolved), conversation_id	Tenant

Catalogue

Table	Key columns	Scope
catalog_items	id, company_id, catalog_type_id, title, status, availability, booking_status, price_paise (bigint), attributes (jsonb, validated by field_definitions), search_tsv, created_at, updated_at	Tenant
catalog_media	id, company_id, item_id, kind (photo / brochure / video), file_key, sort_order	Tenant

Table	Key columns	Scope
catalog_collections · collection_items	id, company_id, catalog_type_id, name, mode (manual / filter), filter (jsonb) · collection_id, item_id	Tenant

CRM

Table	Key columns	Scope
leads	id, company_id, number (per-company sequence, LD-000123), name, phone_e164, email, source_id, status_id, score_label (hot / warm / cold), score_value (0-100), score_reason, assigned_ai_employee_id, assigned_user_id, preferred_language, tags (text[]), custom (jsonb), consent_status, dnd, follow_up_at, last_contact_at, next_action, created_via, created_at, updated_at, deleted_at. Unique (company_id, phone_e164); GIN on custom	Tenant
lead_activities	id, company_id, lead_id, type (call / whatsapp / chat / email / note / status_change / score_change / appointment / assignment / handoff / import), ref_type, ref_id, summary, actor_type, actor_id, created_at. Monthly partitions	Tenant
lead_notes	id, company_id, lead_id, author_type, author_id, body, created_at	Tenant
lead_score_events	id, company_id, lead_id, label, value, reason, rule_hits (jsonb), source (rules / ai), created_at	Tenant
scoring_rules	id, template_version_id | company_id, name, condition (jsonb), points, force_label, priority, active	Platform / Tenant
tasks	id, company_id, lead_id, assigned_user_id, type (callback / follow_up / other), title, due_at, status, created_by_type, created_by_id	Tenant
assignment_rules	id, company_id, name, condition (jsonb), strategy (round_robin / user / team), targets (uuid[]), priority, active	Tenant
consent_records	id, company_id, lead_id, channel, action (opt_in / opt_out), source, evidence (jsonb), created_at	Tenant
lead_imports	id, company_id, file_key, mapping (jsonb), status, total, created, updated, skipped, errors_file_key, created_by	Tenant

Calendar

Table	Key columns	Scope
calendar_resources	id, company_id, kind (user / site / room / other), user_id, name, timezone, working_hours (jsonb), slot_minutes, buffer_minutes, max_per_slot, active	Tenant
availability_exceptions	id, company_id, resource_id, starts_at, ends_at, kind (leave / blocked / extra)	Tenant
holidays	id, company_id (null = platform), date, name, region	Platform / Tenant
appointments	id, company_id, lead_id, appointment_type_id, resource_id, starts_at, ends_at, status (scheduled / confirmed / rescheduled / cancelled / completed / no_show), location, meeting_link, notes, created_by_type, created_by_id, channel, conversation_id, reminders (jsonb). EXCLUDE constraint prevents overlapping bookings per resource	Tenant

Conversations & Channels

Table	Key columns	Scope
conversations	id, company_id, ai_employee_id, lead_id, channel (voice / whatsapp / web_chat), direction, status, handled_by (ai / human / mixed), campaign_id, started_at, ended_at, language, summary, requirement (jsonb), sentiment, intent, intent_confidence, outcome, follow_up_required, follow_up_at, transfer_status (none / requested / completed / failed / no_agent), score_label_after	Tenant
messages	id, company_id, conversation_id, role (customer / ai / human / system / tool), content, language, media (jsonb), start_ms, end_ms, confidence, provider_message_id, delivery_status, citations (jsonb), created_at. Monthly partitions; holds call transcripts too	Tenant

Table	Key columns	Scope
calls	id, company_id, conversation_id, phone_number_id, provider_id, provider_call_id, from_e164, to_e164, status, queued_at, answered_at, ended_at, duration_sec, billable_sec, hangup_by, amd_result, recording_key, recording_status, latency (jsonb per stage), cost_paise	Tenant
tool_invocations	id, company_id, conversation_id, tool_key, input (jsonb), output (jsonb), status (ok / error / denied / timeout), latency_ms, idempotency_key, created_at	Tenant
handoffs	id, company_id, conversation_id, reason, requested_at, assigned_user_id, accepted_at, ended_at, status (pending / accepted / missed / no_agent / completed)	Tenant
phone_numbers	id, company_id, provider_id, e164, capabilities, ai_employee_id, is_test, status	Tenant
whatsapp_accounts	id, company_id, provider_id, waba_id, phone_number_id, display_number, ai_employee_id, credentials_enc, quality_rating, status	Tenant
whatsapp_templates	id, company_id, whatsapp_account_id, name, language, category, components (jsonb), provider_status (pending / approved / rejected / paused), provider_template_id, rejection_reason	Tenant
web_widgets	id, company_id, ai_employee_id, public_key, allowed_domains (text[]), appearance (jsonb), status	Tenant

Campaigns

Table	Key columns	Scope
campaigns	id, company_id, name, description, ai_employee_id, status (draft / scheduled / running / paused / completed / stopped), audience (jsonb), collection_id, schedule (jsonb: dates, days, windows, timezone), retry_rules (jsonb), follow_up_rules (jsonb), max_concurrency, daily_limit, approved_by, started_at, completed_at, stats (jsonb)	Tenant
campaign_targets	id, company_id, campaign_id, lead_id, status (pending / in_progress / connected / completed / failed / skipped_dnd / skipped_opt_out / exhausted), attempts, last_attempt_at, next_attempt_at, last_outcome, conversation_id. Index (campaign_id, status, next_attempt_at)	Tenant

Billing & Usage

Table	Key columns	Scope
plans	id, key, name, description, price_monthly_paise, price_yearly_paise, gst_rate, trial_days, is_public, is_custom, razorpay_plan_ids (jsonb), status, sort_order	Platform
plan_entitlements	plan_id, key (ai_employees / users / voice_minutes / calls / whatsapp_conversations / ai_credits / kb_storage_mb / concurrent_calls / campaigns / feature:), limit_value, overage_price_paise, enforcement (hard / soft)	Platform
subscriptions	id, company_id, plan_id, status (trialing / active / past_due / cancelled / expired), razorpay_subscription_id, current_period_start, current_period_end, cancel_at, entitlement_overrides (jsonb)	Tenant
payments	id, company_id, subscription_id, razorpay_payment_id, razorpay_order_id, amount_paise, status, method, paid_at, raw (jsonb)	Tenant
invoices	id, company_id, number (series per financial year), period_start, period_end, lines (jsonb), subtotal_paise, cgst_paise, sgst_paise, igst_paise, total_paise, place_of_supply, status, pdf_key, issued_at	Tenant
usage_events	id, company_id, ai_employee_id, conversation_id, meter, quantity, unit, provider_id, provider_cost_paise, occurred_at, idempotency_key (unique). Append-only, monthly partitions	Tenant
usage_daily	company_id, date, meter, quantity, provider_cost_paise (rollup)	Tenant
credit_wallets · credit_transactions	company_id, balance · id, company_id, kind (grant / top_up / consume / expire / adjust), amount, reference, created_by, created_at	Tenant

Operations

Table	Key columns	Scope
audit_logs	id, company_id (nullable), actor_type, actor_id, action, entity_type, entity_id, changes (jsonb), ip, user_agent, request_id, created_at. Append-only (no UPDATE / DELETE grant), monthly partitions	Both
notifications	id, company_id, user_id, type, title, body, link, read_at, created_at	Tenant
webhook_events	id, provider, event_type, external_id (unique per provider), signature_valid, payload (jsonb), status, attempts, processed_at	Platform
support_tickets · support_messages	id, company_id, created_by, subject, priority, status, assignee_staff_id · ticket_id, author, body, attachments	Tenant
platform_enquiries	id, name, business, phone, email, industry, message, status, created_at	Platform
data_requests	id, company_id, kind (export / delete), status, requested_by, scheduled_for, completed_at, file_key	Tenant

Post-MVP (Phase 9)

Table	Key columns	Scope
support_cases	id, company_id, number, lead_id, conversation_id, subject, description, channel, priority, status (open / in_progress / waiting / resolved / closed), assigned_user_id, created_by_type, timestamps	Tenant
receivables	id, company_id, lead_id, reference, amount_paise, due_date, status (due / reminded / promised / paid / escalated), reminders_sent, promise_date, escalated_at	Tenant
external_connectors	id, company_id, name, base_url, auth_type, credentials_enc, endpoints (jsonb), allowed_hosts, timeout_ms, status	Tenant

15.4 Indexing, Partitioning and Retention

- B-tree indexes on (company_id, …) for every list and filter path; GIN on JSONB custom fields and tsvector columns; HNSW on embeddings.
- Partitioned tables keep hot data small; old partitions archived to object storage per retention settings.
- Retention jobs: recordings and transcripts per company setting; audit logs per platform policy; usage events retained for billing and tax periods.
- Analytics queries run on the read replica; dashboard KPIs use pre-aggregated rollups.

16. API Architecture

16.1 Principles

- REST over HTTPS with JSON; versioned paths (/api/v1); OpenAPI schema generated from code (drf-spectacular) and used to generate the Dart client for the future Flutter app.
- Authentication: Django session (django-allauth with MFA) in httpOnly, Secure, SameSite cookies for the web apps; CSRF protection on state-changing requests.
- Authorisation on every endpoint: role permission + object-level tenant check.
- Cursor pagination, filtering and sorting via query parameters; consistent error format; request ids.
- Idempotency-Key header on POSTs that create payments, bookings, calls or messages.
- Rate limits per IP, user, company, widget key and webhook source (Redis token buckets).

16.2 API-First & Mobile App Readiness

Every section of the platform is built in Django **together with its REST APIs, in the same phase**. The web pages contain no business logic of their own; page views and APIs call the same service layer. A future **Flutter mobile app (Android + iOS)** therefore connects to the same system, sees the same data and runs the same workflows; only the mobile design and app-side code need to be written.

Concern	How the API is made mobile-ready
Coverage	Each feature's definition of done includes its endpoints, serializers, permissions, OpenAPI docs and API tests.
Authentication	Same /auth endpoints. Web: Django session cookies. Mobile: Bearer access token (short-lived) + rotating refresh token returned in the body, stored in iOS Keychain / Android Keystore; refresh-token reuse detection revokes the session.
Same rules	Identical RBAC, company isolation, rate limits and audit logging for web and mobile clients.
Client generation	OpenAPI schema generates the Dart client for Flutter (openapi-generator, dart-dio) and the API reference for our own JavaScript.
Versioning	/api/v1 kept backward-compatible; breaking changes only in /api/v2 so released app versions keep working; minimum supported app version returned by a config endpoint.
Push notifications	POST /devices registers Firebase Cloud Messaging tokens (Android and iOS); notifications service sends hot-lead, transfer, appointment and inbox alerts.
Files & real-time	Uploads / downloads via pre-signed URLs; the same WebSocket channel delivers live events to apps.
Payloads	Cursor pagination, field filtering and compact responses suited to mobile networks.

16.3 API Surfaces

Surface	Base path / host	Consumers	Auth
Company API	/api/v1	Company panel, mobile PWA, future Flutter app	Cookie (web) or Bearer token (mobile) + membership + role
Admin API	/admin-api/v1 on the admin host	Super Admin app	Staff session, 2FA, IP allowlist
Public API	/public/v1	Website, registration, widget	Anonymous / widget key, rate limited
Webhooks	/webhooks/{provider}	Telephony, WhatsApp, Razorpay, email	Provider signature verification
Realtime	wss /ws/app, /ws/widget	Panels (live updates, inbox), widget	Session / widget session
Voice media	wss voice host /media/{provider}	Telephony provider media streams	Signed one-time stream token
Internal	/internal/v1	Voice Gateway, workers	Service token over private network

16.4 Endpoint Catalogue

Area	Endpoints
Auth	POST /auth/register · /auth/login · /auth/refresh · /auth/logout · /auth/verify-email · /auth/otp/send · /auth/otp/verify · /auth/password/forgot · /auth/password/reset · /auth/mfa/setup · /auth/mfa/verify · GET /auth/sessions · DELETE /auth/sessions/{id}
Company & team	GET / PATCH /company · GET /me · PATCH /me · PATCH /me/availability · GET / POST /team/members · PATCH / DELETE /team/members/{id} · POST /team/invitations · POST /team/invitations/{id}/resend
AI Employees	GET / POST /ai-employees · GET / PATCH / DELETE /ai-employees/{id} · POST /ai-employees/{id}/activate · /pause · GET /ai-employees/{id}/versions · POST /ai-employees/{id}/test-sessions · GET /ai-employees/{id}/analytics · GET /roles · GET /voices?language=
Knowledge	POST /kb/uploads (pre-signed URL) · GET / POST /kb/sources · POST /kb/sources/{id}/approve · /reprocess · DELETE /kb/sources/{id} · GET / POST / PATCH /kb/faqs · POST /kb/test-query · GET /kb/unanswered
Catalogue	GET /catalog-types · GET / POST /catalog/{type}/items · GET / PATCH / DELETE /catalog/items/{id} · POST /catalog/{type}/import · GET / POST /catalog/collections
CRM	GET / POST /leads · GET / PATCH / DELETE /leads/{id} · GET /leads/{id}/timeline · POST /leads/{id}/notes · POST /leads/bulk · POST /leads/import · GET /leads/imports/{id} · POST /leads/export · POST /leads/merge · GET /lead-fields · GET /lead-statuses · GET / POST /tasks · GET / POST /assignment-rules · GET / PATCH /scoring-rules
Calendar	GET / POST /calendar/resources · GET /calendar/availability · GET / POST /appointments · PATCH /appointments/{id} · POST /appointments/{id}/cancel
Conversations	GET /conversations · GET /conversations/{id} · GET /conversations/{id}/messages · POST /conversations/{id}/takeover · /handback · POST /conversations/{id}/messages · GET /calls/{id}/recording-url · POST /calls (call now) · GET /handoffs/pending · POST /handoffs/{id}/accept
Campaigns	GET / POST /campaigns · GET / PATCH /campaigns/{id} · POST /campaigns/{id}/start · /pause · /resume · /stop · GET /campaigns/{id}/targets · GET /campaigns/{id}/stats
Channels	GET / POST /phone-numbers · GET / POST /whatsapp/accounts · GET / POST /whatsapp/templates · POST /whatsapp/templates/{id}/sync · GET / POST / PATCH /widgets
Analytics	GET /analytics/overview · GET /analytics/timeseries?metric=&from=&to=&employee= · GET /analytics/campaigns/{id}
Billing	GET /billing/plans · GET /billing/subscription · POST /billing/checkout · POST /billing/change-plan · POST /billing/cancel · GET /billing/invoices · GET /billing/usage · GET /billing/credits · POST /billing/top-up
Other	GET /notifications · POST /notifications/read · GET / POST /support/tickets · POST /data-requests
Mobile	POST /devices (register push token) · DELETE /devices/{id} · GET /app-config (minimum app version, feature flags) · login / refresh return Bearer tokens when called by the mobile client
Admin API	/companies (+ suspend, activate, impersonate, plan) · /users (+ lock, reset-2fa) · /industries · /industries/{id}/versions (+ publish, rollback, clone) · /field-definitions · /lead-statuses · /lead-sources · /scoring-rules · /catalog-types · /appointment-types · /holidays · /roles · /tools · /languages · /voices · /providers (+ test, health) · /model-routes · /plans · /plan-entitlements · /subscriptions · /payments · /invoices · /credits/adjust · /usage · /calls · /ai-employees · /settings · /settings/general (+ logo / favicon upload) · /feature-flags · /cms/pages · /cms/sections (+ reorder, show / hide, publish) · /cms/media · /notification-templates · /audit-logs · /system-logs · /webhook-events · /support/tickets · /enquiries · /data-requests · /staff
Public API	GET /public/v1/branding · GET /public/v1/plans · GET /public/v1/industries · GET /public/v1/cms/{page} · POST /public/v1/enquiries · GET /public/v1/widget/{key}/config · POST /public/v1/widget/{key}/sessions
Webhooks	POST /webhooks/telephony/{provider} · GET / POST /webhooks/whatsapp/{provider} · POST /webhooks/razorpay · POST /webhooks/email/{provider}
Internal	POST /internal/v1/calls/{id}/context · POST /internal/v1/tools/execute · POST /internal/v1/calls/{id}/events · POST /internal/v1/calls/{id}/end

16.5 Error Format

```
HTTP/1.1 402 Payment Required
{
  "error": {
    "code": "plan_limit_reached",
    "message": "Your plan allows 3 AI Employees.",
    "details": {"limit": 3, "current": 3, "entitlement": "ai_employees"},
    "request_id": "req_01JA4Z7W3K9X"
  }
}
```

17. Frontend Architecture

No separate JavaScript framework (Next.js / React) is used. The website and both panels are **Django templates** built directly on the purchased Bootstrap / jQuery templates, which already ship as Django projects. Pages are rendered on the server by Django, so SEO works and there is one language, one project and one deployment.

Surface	Template (in project folder)	How it is built
Public website	WebsiteTemplate / eSoft: HR layout (index5), light theme	Django templates + Bootstrap 5 + jQuery; GSAP, ScrollTrigger, AOS animations kept as they are; every section's content comes from Super Admin (Section 6)
Super Admin	AdminTemplate / Rizz, light theme	Django templates + Bootstrap 5.3; own sidebar menu; served on the admin sub-domain
Company panel	AdminTemplate / Rizz (same template)	Same templates and components, company sidebar and accent colour
Chat widget	Built by us	Small vanilla JavaScript bundle in an iframe (≤ 40 KB gzip), loaded from the CDN
Mobile app (future)	Flutter	Uses the same REST APIs

```
backend/
  templates/website/          eSoft templates (HR layout); base, partials (header, footer), sections/
  templates/panel/            Rizz templates shared by Super Admin and company panel; partials, components/
  templates/panel/admin/      Super Admin pages (own sidebar)
  templates/panel/company/    company panel pages (own sidebar)
  static/website/             eSoft CSS, JS, fonts, images
  static/panel/               Rizz compiled CSS / JS (built once from SCSS with gulp, then committed)
  static/widget/              chat widget bundle
```

- **Service layer rule:** all business logic lives in Python services. Django page views and the REST API both call the same services, so nothing is written twice and the API stays complete for the Flutter app.
- **Interactive screens:** jQuery + AJAX calls to our own REST APIs, WebSockets (Django Channels) for live inbox, handoff screen pops and live dashboards. If a screen becomes too complex, a small library (HTMX or Alpine.js) is added only for that screen; no rewrite needed.
- **Rizz pages reused:** calendar → appointments, chat → WhatsApp / chat inbox, kanban → lead pipeline, form wizard → AI Employee wizard, datatables → leads / calls / companies, contact list → leads, invoice → GST invoices, pricing → plans, notifications, analytics → dashboards, auth pages → login / 2FA / recovery.
- **Dynamic UI:** lead and catalogue forms, tables and filters are rendered from field definitions by shared template components; nothing industry-specific is hard-coded.
- **Authentication:** django-allauth (already used by both templates) with MFA for web login; the REST API adds Bearer tokens for mobile.
- **Light theme** by default for website and panels (decided); Rizz also supports dark mode if ever needed.
- **Not copied into our project:** template Docs folders, demo databases, RTL and extra locales, unused demo home pages and UI-kit showcase pages.
- **Mobile:** responsive Bootstrap layouts, tables collapse to cards, installable PWA with push notifications.
- **Quality:** djLint for templates, ESLint for our own JavaScript, Playwright E2E, accessibility checks (axe).
- **Future Flutter app:** Android and iOS app built on the same Company API with a generated Dart client; no backend changes needed because every feature already has its REST API.

18. Security Architecture

18.1 Identity and Access

- Passwords hashed with Argon2; configurable password policy; breach-safe reset tokens (single use, short expiry).
- TOTP 2FA: mandatory for staff, optional / enforceable for companies; lockout and rate limiting on auth endpoints.
- RBAC with permission keys; object-level tenant checks; least privilege for staff (Super Admin vs Support).
- Mobile clients: short-lived Bearer tokens, rotating refresh tokens with reuse detection, per-device sessions that users or admins can revoke.
- Support impersonation requires a reason, shows a banner and is fully audit-logged.

18.2 Data Protection

- TLS 1.2+ everywhere, HSTS; encryption at rest for database, files and backups.
- Envelope encryption for provider credentials, WhatsApp tokens and 2FA secrets (*_enc columns).
- Infrastructure secrets (SECRET_KEY, database URL) only in the server environment or the host's secret store; provider API keys only in Super Admin (encrypted). Never in code, images or the repository (secret scanning in CI).
- Private file storage; downloads via signed URLs valid for minutes; uploads type-checked, size-limited, malware-scanned.
- Personal data masked in logs and traces; data export and deletion workflows (TEN-07).

18.3 Network and Application Security

- Services in private subnets; only the load balancer is public; database reachable only from application security groups.
- Web application firewall / rate limiting at the reverse proxy (or the host's WAF); Super Admin restricted by IP allowlist.
- Security headers (CSP, X-Frame-Options except widget iframe, Referrer-Policy); CSRF protection; strict CORS.
- Webhooks: signature verification, timestamp tolerance, replay protection via event id de-duplication.
- Outbound HTTP (crawler, connectors) through an egress guard blocking private IP ranges (SSRF protection).

18.4 Threat Model Summary

Threat	Mitigation
Cross-tenant data access	Tenant resolution from trusted source, ORM scoping, RLS, isolation test suite
Account takeover	2FA, lockout, rate limits, session revocation, login alerts
Prompt injection / AI misuse	Guardrail precedence, tool scoping, server-side company_id, evaluation gate (Section 8)
Toll fraud / call pumping	Outbound only to leads, +91 default allowlist, per-company daily caps, anomaly alerts
Webhook spoofing	Signatures, replay protection, provider IP allowlists where available
Malicious uploads	Type sniffing, size limits, malware scan, never served inline
SSRF via crawler / connectors	Egress guard, private-range blocking, timeouts, host allowlists
Insider misuse	Staff RBAC, audit logs, impersonation reason + banner, least privilege DB roles
Dependency vulnerabilities	Dependabot, pip-audit, npm audit, Trivy in CI; patch SLAs

18.5 Audit and Compliance Controls

- Append-only audit_logs for configuration changes, data exports, recording access, impersonation, billing actions.
- Consent records, opt-out handling, calling windows, attempt caps and AI disclosure are enforced by the platform (PRD-01 M26), configurable from Super Admin.
- External penetration test before launch; DAST on staging at every gate; quarterly access reviews after launch.

19. Infrastructure & Deployment

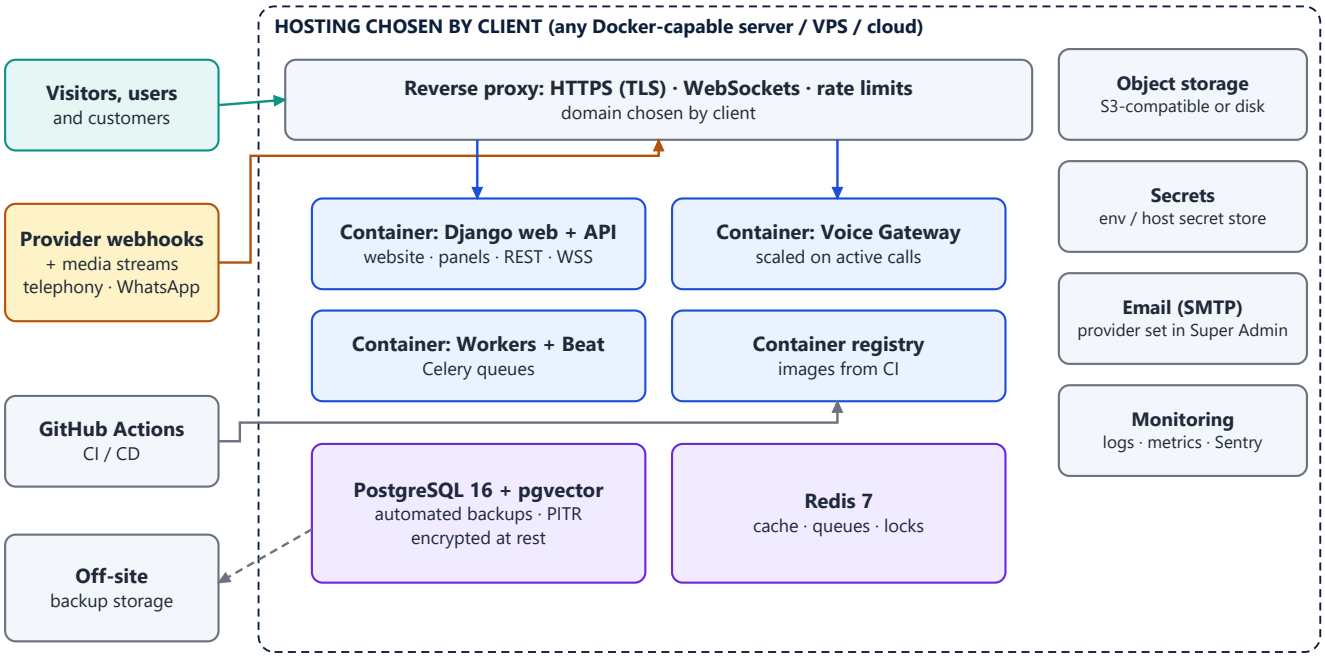


Figure 10: Deployment (hosting chosen by the client; any Docker-capable server)

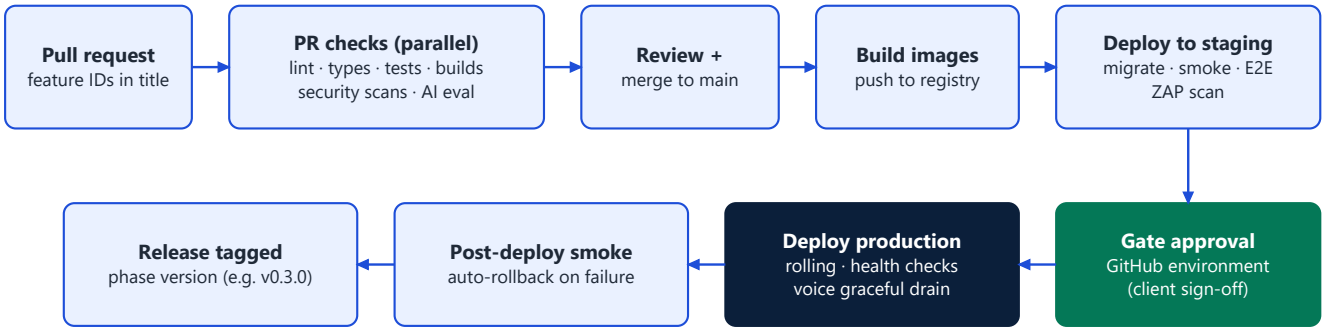
19.1 Environments

Environment	Purpose	Providers
Local	Developer machines: Docker Compose with PostgreSQL + pgvector, Redis, Mailpit, FreeLLMAPI	Mocks, sandbox keys, free LLM tiers via FreeLLMAPI
Staging	Production-like test server (chosen with the client); phase demos, UAT, DAST	Test telephony account, Razorpay test mode, WhatsApp test number, FreeLLMAPI (real model for gate tests)
Production	Live customers	Client's own telephony and WhatsApp accounts, Razorpay live

19.2 Deployment Process

- Hosting and domain are decided by the client before launch (Phase 8); until then nothing provider-specific is built.
- Container images built once, promoted from staging to production by digest.
- Database migrations run as a one-off task before rollout; always backward-compatible.
- Rolling container restarts with health checks and rollback to the previous image; Voice Gateway drains active calls.
- Feature flags for risky features; production deploy requires the GitHub environment approval (phase gate).
- Portability: Docker images + compose files + provider adapters run on any VPS or cloud the client chooses.

20. CI/CD & Phase Gates



A failed check anywhere stops the pipeline. Production deploys only from a green pipeline and an approved gate.

Figure 11: Delivery pipeline from pull request to production

Required check	Tooling	Runs on
Lint and format	ruff, black, djLint (templates), eslint (our JS)	Every PR
Type checks	mypy	Every PR
Unit and integration tests + coverage ≥ 80 % (backend)	pytest (PostgreSQL + Redis services)	Every PR
Migration consistency	makemigrations --check, migration lint	Every PR
Builds	collectstatic, widget size budget, Docker builds	Every PR
Tenant isolation suite	pytest cross-tenant tests	Every PR
AI evaluation	Golden conversations per template × language	PRs touching AI + nightly
Security	Bandit, Semgrep, pip-audit, npm audit, Gitleaks, Trivy	Every PR
E2E	Playwright against staging	Every staging deploy
DAST	OWASP ZAP baseline	Every staging deploy

Gate rule

A phase exits only when the release tag has a green pipeline, security checks show no open critical or high findings, the no-hard-coding review is signed, staging UAT is done and the client approves (PRD-01 Section 4).

21. Observability

Signal	What we collect	Tooling
Logs	Structured JSON with request_id, company_id, conversation_id, call_id; personal data masked	Per hosting (container logs, Loki or host logging)
Metrics	API latency / errors, queue depth, active calls, per-stage voice latency, STT confidence, interruptions, tool error rate, tokens and cost per company, webhook failures	Prometheus-compatible
Traces	End-to-end traces across API, workers and Voice Gateway, including provider calls	OpenTelemetry
Errors	Exceptions with context and release version	Sentry
Dashboards	Platform health, voice quality, provider health, business KPIs	Grafana
Alerts	SLO burn, voice latency P95, provider error spikes, queue backlog, payment webhook failures, unusual outbound volume	Email / WhatsApp / on-call

22. Reliability & Disaster Recovery

22.1 Service Level Objectives

SLO	Target
API and panels availability	99.9 % monthly
Voice call setup success	≥ 99.5 %
Voice turn latency	P50 ≤ 1.2 s, P95 ≤ 2.0 s
Webhook processing	P95 ≤ 5 s
Knowledge ingestion	100-page PDF ready ≤ 10 minutes
Recovery	RPO ≤ 5 min, RTO ≤ 4 h (region loss)

22.2 Failure Modes

Failure	Behaviour
LLM provider error / slow	Next route in fallback order; if all fail: polite message + human transfer or callback
STT or TTS provider error	Switch to secondary route for the language; alert
Telephony provider outage	Alert; inbound numbers forward to the company's fallback number (provider-side setting)
Database primary failure	Restore from automated backups / promote a replica, depending on the client's hosting setup
Redis failure	Multi-AZ failover; tasks are idempotent and acknowledged late, so they re-run safely
Worker crash	Task retried with exponential backoff; dead-letter queue after max retries
Webhook burst	Fast 200 acknowledgement, durable storage in webhook_events, processing from queue
Region loss	Restore from off-site backups on a new server following the DR runbook

22.3 Backups

- Automated PostgreSQL backups with point-in-time recovery; daily copies stored off-site.
- File storage backed up with versioning where the host supports it; retention rules applied.
- Restore drill at P8 and quarterly after launch; results recorded.

23. Scalability & Capacity

Dimension	Launch design target	Growth target	How it scales
Companies	500	10,000+	Shared schema, indexes led by company_id
AI Employees	2,000	50,000+	Configuration rows; cached effective config
Concurrent calls	100	1,000+	Voice Gateway horizontal autoscaling; provider capacity planning
Calls per month	50,000	1,000,000+	Partitioned messages; async post-processing
WhatsApp messages / month	1,000,000	20,000,000+	Webhook queue; partitioned messages
Knowledge chunks	5 million	100 million+	HNSW + company filter; vector store can move behind the same interface

Targets are design assumptions to be confirmed with the client's business plan in Phase 0.

23.1 Bottlenecks and Mitigations

- **Database writes** (messages, usage, activities): partitioning, batched inserts, rollups, read replica for analytics.
- **Vector search at scale:** HNSW with company filter; if needed, partition by company group or move vectors to a dedicated store behind the retrieval interface.
- **Provider rate limits:** per-provider token buckets, multiple accounts / regions, fallback routes.
- **Queue contention:** separate Celery queues per workload (ingest, post-call, campaigns, notifications, billing).
- **Campaign spikes:** pacing by concurrency limits per campaign and per company.

24. Testing Strategy

Level	Scope	Tools	When
Unit	Services, rules engine, normalisers, adapters (mocked)	pytest	Every PR
Integration	Django apps with real PostgreSQL / Redis	pytest + service containers	Every PR
API contract	OpenAPI schema validity and breaking-change check	drf-spectacular, schema diff	Every PR
Tenant isolation	Every endpoint, tool and retrieval cross-tenant	pytest suite	Every PR
E2E	Registration, wizard, CRM, calendar, inbox, billing, Super Admin	Playwright	Staging deploys
AI evaluation	Grounding, extraction, tool use, handoff per template × language	Evaluation harness	AI changes + nightly
Voice	Scripted calls (recorded Hindi / English / Bengali audio), latency, barge-in, transfer	Call simulator + test numbers	P5 onward
Load	API throughput, concurrent calls, campaign pacing	k6, call simulator	P5, P8
Security	SAST, dependency, secrets, container, DAST, pen-test	Bandit, Semgrep, ZAP, external firm	PRs / gates / P8
Accessibility & devices	axe checks; desktop, laptop, tablet, Android, iPhone	axe, BrowserStack / devices	Every gate
UAT	Client test cases per phase	UAT sheets on staging	Every gate

25. Repository Structure & Engineering Standards

```
ai-employee/
  backend/
    config/                 settings (12-factor), urls, asgi, celery
    apps/                   accounts, tenancy, rbac, platform_config, templates_engine, roles_tools,
                            ai_employees, knowledge, catalog, crm, scoring, calendar_app, conversations,
                            voice, messaging, widget, handoff, campaigns, analytics, billing, usage,
                            notifications, support, audit, cms, compliance
    packages/
      agent_runtime/        channel-independent AI engine
      providers/            llm/ stt/ tts/ telephony/ whatsapp/ email/ sms/ payment/ embeddings/
      tenancy/              tenant context, managers, RLS helpers
    tests/                  isolation suite, AI evaluation sets, fixtures
  voice_gateway/            asyncio service (Pipecat), imports backend/packages
  backend/templates|static  website (eSoft) and panels (Rizz), see Frontend Architecture
  mobile/                   (future) Flutter app for Android + iOS, uses the same /api/v1
  deploy/                   production compose files (Phase 8, for the hosting the client chooses)
  .github/workflows/        ci.yml, deploy-staging.yml, deploy-production.yml, nightly-ai-eval.yml
  docs/                     PRD, SDD, ADRs, runbooks
```

- Python 3.12+, Django 5.2 LTS; pre-commit hooks for formatting and linting.
- Short-lived feature branches, pull requests with feature IDs (e.g., "CRM-04: dynamic fields"), at least one reviewer, CODEOWNERS for sensitive areas (tenancy, billing, security).
- Conventional commits; release tags per phase (v0.1.0 = P1 … v1.0.0 = MVP launch).
- Architecture Decision Records (docs/adr) for every significant decision.
- No business constants in code: reviewers check the no-hard-coding rule on every PR.

26. Risks & Mitigations

Risk	Impact	Mitigation
Voice latency above target	Unnatural calls	P0 PoC, streaming everywhere, India-region hosting, fast dialogue model, filler phrases
Code-mixed / Bengali recognition accuracy	Wrong extraction	Per-language provider choice, custom vocabulary, evaluation sets, confirmation questions for key fields
AI states wrong facts	Customer trust, liability	Facts only via tools / approved knowledge, numeric checks, zero-tolerance evaluation gate
Provider price or policy change	Cost / outage	Adapters, fallback routes, cost tracking per provider
WhatsApp template approval delays	Feature delay	Prepare templates early in P6; monitor quality rating
Client telephony / WhatsApp onboarding time (KYC, registrations)	Launch delay	Start the client's account paperwork during P5
Running cost per minute above plan margin	Profitability	Usage metering, cost dashboards, plan limits, model routing
Campaign concurrency spikes	Degraded calls	Pacing, per-company limits, autoscaling, load tests
Scope growth	Timeline	Phase gates, change requests, pending list (e.g., AI Marketing Employee)
Security incident	Data exposure	Defence in depth, pen-test, monitoring, incident runbook

27. Open Items

Item	Owner	Resolved in
AI Marketing Employee definition	Client	Pending
Final STT / TTS / LLM providers per language	Development team	P0 PoC
Telephony provider and number plan	Client + development team	P0 / P5
Capacity targets from business plan	Client	P0
Travel package and Education course fields	Client + development team	P3
Details of the eight post-MVP industries	Client + development team	P9 start
Infrastructure sizing and running cost	Development team	Commercial proposal
Flutter mobile app scope and timeline (APIs ready from P1)	Client	Future project

28. Glossary

Term	Meaning
ADR	Architecture Decision Record
ASGI	Asynchronous Server Gateway Interface (async Django)
Barge-in	Caller interrupts while the AI is speaking
BSP	WhatsApp Business Solution Provider
DAST / SAST	Dynamic / static application security testing
DND	Do Not Disturb preference / registry
E.164	International phone number format, e.g. +919876543210
HNSW	Graph index for fast vector search
LLM	Large language model
PITR	Point-in-time recovery
PWA	Progressive Web App (installable web app)
RAG	Retrieval-augmented generation: answering from retrieved approved knowledge
RLS	PostgreSQL row-level security
RPO / RTO	Maximum data loss / maximum time to restore
SLO	Service level objective
STT / TTS	Speech-to-text / text-to-speech
TTFT / TTFB	Time to first token / first audio byte
VAD	Voice activity detection
WABA	WhatsApp Business Account